

TECHNICAL BULLETIN

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PRE-CATALYZED NADIC METHYL ANHYDRIDE (NMA 419)

Typical Properties

Properties	Values	Test Method
Molecular Weight (average)	204	
Specific gravity @ 20°C, g/cc	1.197 – 1.210	D-102 (ASTM D 4052)
Appearance	Dark green liquid	D-104 (Visual)
Viscosity at 25°C, cP	400 - 1000	D-237

Applications

NADIC Methyl Anhydride 419 (NMA 419) is an epoxy curing agent formulated with an efficient latent proprietary catalyst, coupled with an internal mold release agent. This curative blend greatly improves the reactivity of NMA in epoxy curing and provides an outstanding surface finish to pultruded profiles. Although highly reactive at process temperatures, NMA 419 maintains excellent pot life. NMA 419 is commonly used to cure epoxy resins in many challenging applications, including fiber reinforced composites used at high temperatures and those that require resistance to heat cycling. NMA 419 is ideal for use in processes such as pultrusion, filament winding and infusion.

NMA 419 provides the following benefits:

- **Easy Handling:** Low viscosity liquid gives low viscosity mixes with various epoxy resins
- **Convenience:** Eliminates the need to add small amounts of catalyst and mold release

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- **Consistency:** Prevents potential batch-to-batch differences caused by errors in measuring and properly dispersing additives
- **Low reactivity at room temperature:** Provides long working times and pot lives for mixes with epoxy resins
- **High reactivity at cure temperature:** Excellent cure response at cure temperatures, resulting in extensive crosslinking and high Tg
- **High Performance:** High heat distortion temperature (HDT) and glass transition temperature (Tg). Good thermal stability. Improved mechanical properties including hardness and impact strength.
- **High-speed pultrusion process:** Improved surface quality, ease of processing and better line speeds than traditional NMA-cured epoxy pultrusion.

For more information on the use of anhydrides like NMA 419 as epoxy curing agents, please consult the Dixie Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems."

Epoxy Formulations

NMA 419 was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 150°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

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Epoxy type	NMA 419, phr	Viscosity at 25°C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	102	2450	19.4	163
Low Viscosity BPA Liquid	106	1920	20.6	164
Cycloaliphatic Epoxy	141	450	19.7	250
Epoxy Phenol Novolac	112	2020	20.8	158
Epoxy BPA Novolac	101	6720	20.6	177
BPF Liquid Epoxy	114	1160	18.6	156

1. Shyodu Hot Pot (100g at 100°C).

2. Cured 1 hr at 150°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

It is recommended that NMA 419 is stored between the temperatures of 18°C (85°F) and 40°C (104°F) to avoid freezing, although the material is not affected by freezing. If material does freeze, thaw at a temperature below 40°C and mix well before using.

NMA 419 will react with water to form diacids. This is normally undesirable, so NMA 419 should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for anhydrides such as NMA 419, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected

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skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

NMA 419 is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.